Unbalanced Mach–Zehnder Interferometers for Measuring the Group Index of Silicon Strip Waveguides

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***Abstract*—This report describes the design of four unbalanced Mach–Zehnder interferometers (MZIs) and a grating coupler loopback for the openEBL-2026-10 electron beam lithography run. All devices use $500\,\mathrm{nm}$ wide and $220\,\mathrm{nm}$ thick silicon strip waveguides in the fundamental quasi-TE mode near $1550\,\mathrm{nm}$. The MZIs have path length differences of 50, 100, 150 and $200\,\mu\mathrm{m}$. The free spectral range of each MZI will be measured and used to calculate the group index of the fabricated waveguide, which will then be compared with simulation.**

***Index Terms*—Silicon photonics, Mach–Zehnder interferometer, group index, compact model.**

I. Introduction

SILICON photonics uses the large refractive index contrast between silicon ($n \approx 3.48$) and silicon dioxide ($n \approx 1.44$) at $1550\,\mathrm{nm}$ to confine light in waveguides that are only a few hundred nanometres wide. This allows compact optical circuits to be made with the same fabrication tools that are used for CMOS electronics [1]. Mach–Zehnder interferometers (MZIs) are one of the most common building blocks in these circuits. They are used as wavelength filters and multiplexers in optical communication links, as switches and modulators, and as sensors.

The group index n_g of a waveguide is needed to design most interferometers and resonators, because it sets their free spectral range (FSR), for example the channel spacing of a wavelength filter. It can be calculated with a mode solver, but the fabricated waveguide is never exactly the designed one, and small changes in its width and thickness change n_g. It is therefore useful to measure n_g on the fabricated chip.

An unbalanced MZI is one of the simplest ways to do this. Its transmission spectrum is periodic in wavelength, and the period depends only on n_g and on the length difference ΔL between the two arms. Since ΔL is fixed by the layout, measuring the FSR gives n_g.

The objective of this design is to extract n_g of the fabricated waveguide from measured MZI spectra and compare it with the simulated value. For this, I chose four MZIs with $\Delta L = 50$, 100, 150 and $200\,\mu\mathrm{m}$, and a loopback structure made of two grating couplers joined by a straight waveguide. The loopback measures the response of the grating couplers, so that it can be removed from the MZI spectra. Having four values of ΔL gives four separate estimates of n_g from the same chip, which can be compared with each other and with the simulation.

The chip is fabricated by Applied Nanotools on an SOI wafer with a $220\,\mathrm{nm}$ silicon layer, using a single full etch and an oxide cladding [2]. The designs have to fit in a $605\,\mu\mathrm{m} \times 410\,\mu\mathrm{m}$ area. Light is coupled into and out of the chip with TE grating couplers spaced by $127\,\mu\mathrm{m}$, the pitch of the fibre array in the measurement setup. According to the course guidelines, the tunable laser has a minimum step of $1\,\mathrm{pm}$ and the grating couplers limit the usable bandwidth to about $50\,\mathrm{nm}$. The FSR of each MZI must therefore be well below $50\,\mathrm{nm}$, so that several fringes fall inside the measured range.

II. Theory

A. Effective index, group index and compact model

The waveguide is used in its fundamental quasi-TE mode, in which the electric field is mostly parallel to the chip surface. The phase of this mode is described by its effective index n_{eff}. Over a length L the mode picks up a phase βL, where $\beta = 2\pi n_{\mathrm{eff}}/\lambda$ is the propagation constant and λ is the free-space wavelength.

In silicon waveguides n_{eff} changes strongly with wavelength, mainly because the size of the mode changes with wavelength (waveguide dispersion). Over a limited wavelength range it can be written as a second-order polynomial around the centre wavelength $\lambda_0 = 1.55\,\mu\mathrm{m}$ [1]:

$$n_{\mathrm{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \tag{1}$$

where n_1, n_2 and n_3 are found by fitting this expression to mode solver results. Equation (1) is the compact model of the waveguide, and it is used later in the circuit simulations.

The group index gives the speed of a wave packet, $v_g = c/n_g$, and is defined as

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda}. \tag{2}$$

At $\lambda = \lambda_0$, using (1), this becomes $n_g(\lambda_0) = n_1 - \lambda_0 n_2$. Because n_{eff} decreases with wavelength ($n_2 < 0$), n_g is larger than n_{eff}.

B. MZI transfer function

In the MZI, a Y-branch splits the input light equally into two arms of length L_1 and L_2, and a second Y-branch combines the two arms into the output waveguide. Each Y-branch multiplies the field in each arm by $1/\sqrt{2}$, so the output intensity is [1]

$$\frac{I_{\mathrm{out}}}{I_{\mathrm{in}}} = \frac{1}{4}\left| e^{-j\beta L_1 - \alpha L_1/2} + e^{-j\beta L_2 - \alpha L_2/2} \right|^2 \tag{3}$$

where α is the power loss per unit length of the waveguide. If the loss is neglected, (3) becomes

$$\frac{I_{\mathrm{out}}}{I_{\mathrm{in}}} = \frac{1}{2}\left[1 + \cos(\beta\,\Delta L)\right] \tag{4}$$

with $\Delta L = L_2 - L_1$. When the light from the two arms arrives in phase ($\beta\Delta L = 2\pi m$), all of it reaches the output. When it arrives out of phase, the combined field cannot be guided by the output waveguide and is radiated at the second Y-branch, so the transmission drops to zero.

In the measurement, the light also passes through two grating couplers, whose transmission $T_{\mathrm{GC}}(\lambda)$ depends on wavelength. The measured spectrum is therefore $T_{\mathrm{GC}}^2(\lambda)$ multiplied by (3). The loopback structure measures $T_{\mathrm{GC}}^2(\lambda)$ on its own, so it can be subtracted from the MZI spectra in dB.

C. Group index from the free spectral range

The FSR is the wavelength spacing between two neighbouring minima (or maxima) of (4). Between two neighbouring minima at λ_1 and $\lambda_2 = \lambda_1 + \mathrm{FSR}$, the phase difference between the arms changes by 2π:

$$\beta(\lambda_1)\,\Delta L - \beta(\lambda_2)\,\Delta L = 2\pi. \tag{5}$$

Since the FSR is small compared with λ, the change in β can be written using its derivative:

$$\beta(\lambda_1) - \beta(\lambda_2) \approx -\frac{d\beta}{d\lambda}\,\mathrm{FSR}. \tag{6}$$

Differentiating $\beta = 2\pi n_{\mathrm{eff}}/\lambda$ and using (2),

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda}\frac{dn_{\mathrm{eff}}}{d\lambda} - \frac{2\pi n_{\mathrm{eff}}}{\lambda^2} = -\frac{2\pi n_g}{\lambda^2}. \tag{7}$$

Substituting (6) and (7) into (5) gives

$$\mathrm{FSR} = \frac{\lambda^2}{n_g\,\Delta L} \tag{8}$$

and, solving for the group index,

$$n_g = \frac{\lambda^2}{\mathrm{FSR}\cdot\Delta L}. \tag{9}$$

Equation (9) is used to find n_g from both the simulated and the measured spectra, with λ taken as the midpoint between the two minima. It also shows that the FSR is set by the group index and not by the effective index. Using n_{eff} instead would give an FSR that is too large by a factor of n_g/n_{eff}.

III. Fabrication

To be completed.

IV. Experimental Data

To be completed.

V. Analysis

To be completed.

VI. Conclusion

To be completed.

References

[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge, U.K.: Cambridge Univ. Press, 2015.

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